INCH-POUND
A-A-52478A
29 September 2017
SUPERSEDING
A-A-52478
21 October 1993

COMMERCIAL ITEM DESCRIPTION

KINGPIN, FIFTH WHEEL, 2 INCH

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This CID covers requirements for a 2 inch fifth wheel kingpin used in coupling military trailer to a truck-tractor or trailer converter dolly fifth wheel. It also includes the requirements for trailers transported on flat car.

2. SALIENT CHARACTERISTICS

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with manufacturer's specifications/drawings. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see Note 5.5).

2.1.1 Steel forging. The kingpins shall be made of steel forgings of carbon content 0.35% max. with sufficient alloy content to be oil quenched and tempered to 302-363 BHN at the surface.

2.2 Design and construction. Unless otherwise specified herein (see figure 1), the kingpin shall be designed and constructed to the form and dimensions in accordance with SAE J700.

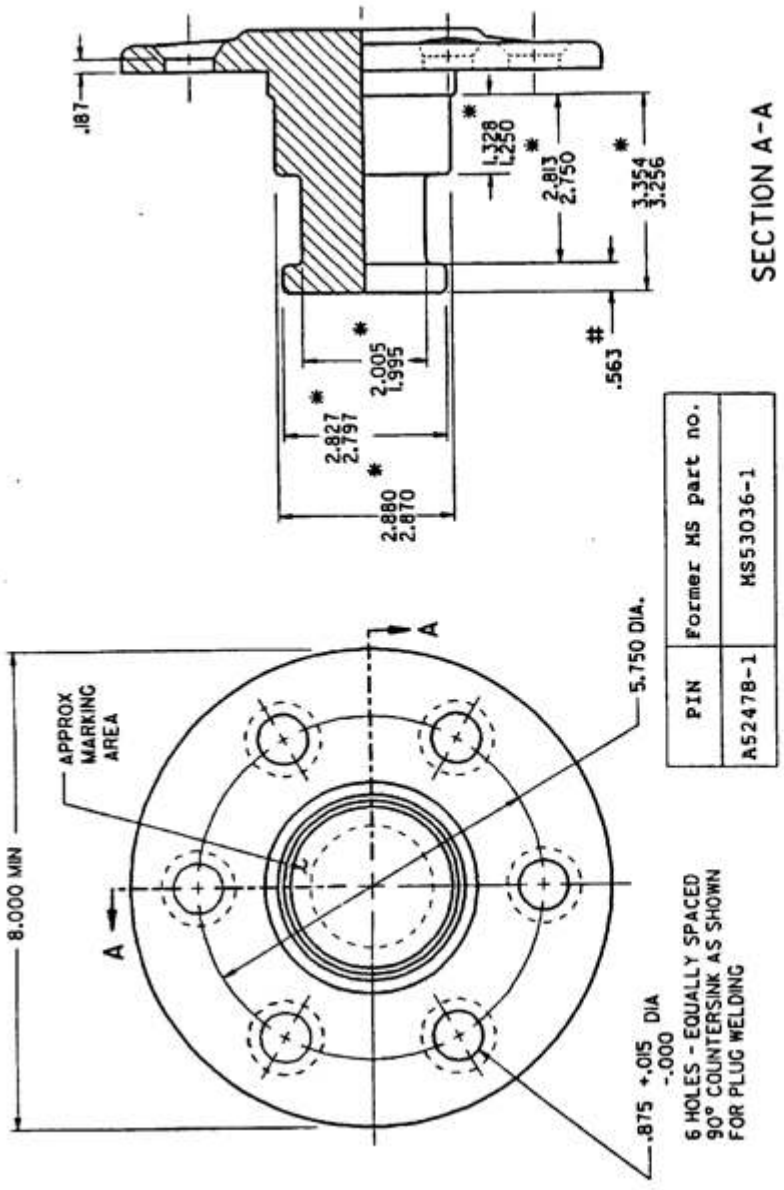
2.3 Performance. When properly installed, the kingpin shall meet the performance requirements in accordance with SAE J133. When transported on a flat car, the kingpin shall withstand force of 3.5 GVW and shall be in accordance with other requirements specified in Association of American Railroads specification M931.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or sent by email to usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 2510

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NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL RADII SHALL BE .125 INCH.
2. DIMENSIONS ARE IN INCHES. DIMENSIONS INDICATED BY POUND SIGN (#) SHALL BE IN ACCORDANCE WITH SAE J700.
3. HEAT TREAT: OIL QUENCH AND TEMPER TO A SURFACE HARDNESS OF 302-363 BHN OVER ALL AREAS OF KINGPIN.
4. THE USE OF ASTERISKS (*) INDICATES THAT DIMENSIONS ARE COVERED BY INTERNATIONAL AGREEMENT (see NOTE 5.7).

FIGURE 1. Kingpin, fifth wheel, 2 inch.

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2.4 Identification markings. Identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code, manufacturer's part number and the part identification number (PIN), A52478-1. The upper coupler plate thickness shall be specified on the kingpin and heat code of lot control for traceability shall be specified (see figure 1 and Note 3.2 Contractor certification).

3. QUALITY ASSURANCE PROVISIONS

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.3 Hardness testing. The kingpin shall receive 100% Brinell Hardness and 100% Magnetic Particle Inspection.

4. PACKAGING

4.1 Packaging. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see note 5.3).

5. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Government documents.

5.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

QSATG-284	Fifth Wheel Attachments
STANAG-4009	Interface Geometry, Tractors and Semitrailers

COMMERCIAL ITEM DESCRIPTIONS

A-A-52542	Fifth Wheel Assembly, 33 Inch Non-Universal
A-A-52544	Fifth Wheel Assembly, 33 Inch Universal

(Copies of these documents are available online at http://quicksearch.dla.mil.)

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ENVIRONMENTAL PROTECTION AGENCY

Public Law 94-580 Resource Conservation and Recovery Act

(Copies of this document are available online at available from www.epa.gov.)

5.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASSOCIATION OF AMERICAN RAILROADS

M931-88 Highway Trailers, All Types, for TOFC Service

(Copies of this document are available online at available from www.aar.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE J133-87 Fifth Wheel Kingpin Performance-Commercial Trailers and Semitrailers

SAE J700-85 Upper Coupler Kingpin-Commercial Trailers and Semitrailers

(Copies of these documents are available online at www.sae.org.)

5.3 Ordering data. Acquisition documents must specify the following:

1. Title, number, and date of this CID.
2. PIN.
3. Selection of applicable level and packaging requirements.

5.4 Associated fifth wheels. See A-A-52542 and A-A-52544 for associated fifth wheels used with this kingpins conforming to this CID.

5.5 Cross-reference data. Kingpins conforming to this CID are interchangeable/substitutable with kingpins conforming to MS53036.

5.6 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

5.7 International standardization agreement. Certain provisions of this specification (see figure 1) are the subject of international standardization agreement QSTAG 284 and NATO-STANAG 4009. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

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MILITARY INTERESTS:

Custodians
Army – AT
Navy – YD
Air Force - 99

Review Activity
Army –CR4, MI
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FSS

PREPARING ACTIVITY:
Army - AT

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